Solution

$$\text{We compute } |\text{adj } A| = \begin{vmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{vmatrix} = 9 \ .$$

$$\text{So, we get } A^{-1} = \pm \frac{1}{\sqrt{|\text{adj}(A)|}} \text{adj}(A) = \pm \frac{1}{\sqrt{9}} \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} = \pm \frac{1}{3} \begin{bmatrix} -1 & 2 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} .$$

Example1.7

If A is symmetric, prove that adj A is also symmetric.

Solution

Suppose A is symmetric. Then, $A^T = A$ and so, by theorem 1.9 (vi), we get

$\text{adj}\left(A^T\right) = \left(\text{adj } A\right)^T \Rightarrow \text{adj } A = \left(\text{adj } A\right)^T \Rightarrow \text{adj } A$ is symmetric.

Theorem 1.10

If A and B are any two non-singular square matrices of order n, then

$$\text{adj}(AB) = (\text{adj } B)(\text{adj } A).$$

Proof

Replacing A by AB in $\text{adj}(A) = |A| A^{-1}$, we get

$$\text{adj}(AB) = |AB|(AB)^{-1} = \left(|B| B^{-1}\right)\left(|A| A^{-1}\right) = \text{adj}(B)\,\text{adj}(A) .$$

Example 1.8

Verify the property $\left(A^T\right)^{-1} = \left(A^{-1}\right)^T$ with $A = \begin{bmatrix} 2 & 9 \\ 1 & 7 \end{bmatrix}$.

Solution

For the given A, we get $|A| = (2)(7) - (9)(1) = 14 - 9 = 5$. So, $A^{-1} = \frac{1}{5}\begin{bmatrix} 7 & -9 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} \frac{7}{5} & -\frac{9}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{bmatrix}$.

$$\text{Then, } \left(A^{-1}\right)^T = \begin{bmatrix} \frac{7}{5} & -\frac{1}{5} \\ -\frac{9}{5} & \frac{2}{5} \end{bmatrix} = \frac{1}{5}\begin{bmatrix} 7 & -1 \\ -9 & 2 \end{bmatrix} . \qquad \ldots (1)$$

For the given A, We get $A^T = \begin{bmatrix} 2 & 1 \\ 9 & 7 \end{bmatrix}$. So $\left|A^T\right| = (2)(7) - (1)(9) = 5$.

$$\text{Then, } \left(A^T\right)^{-1} = \frac{1}{5}\begin{bmatrix} 7 & -1 \\ -9 & 2 \end{bmatrix} . \qquad \ldots (2)$$

From (1) and (2), we get $\left(A^{-1}\right)^T = \left(A^T\right)^{-1}$. Thus, we have verified the given property.

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